

THEORY AND METHODS

A KERNEL METHOD FOR MODELLING INTERVAL CENSORED COMPETING RISKS

Peter Adamic

Laurentian University, Canada

e-mail: padamic@cs.laurentian.ca

and

Sarah Ouadah

Université Pierre et Marie Curie, Paris VI, France

Key words: Density estimation; innermost interval; kernel smoothing; mean squared error; nonparametric maximum likelihood; self-consistency; trypanosomiasis.

Summary: Self-consistent algorithms for modelling the cause-specific cumulative incidence functions in an interval censored, competing risks context were proposed by Hudgens *et al.* (2001) and Adamic (2006). One of the drawbacks of these methods was that convergence was only achieved over a small class of points, called cause-specific innermost intervals. Braun *et al.* (2005) proposed a self-consistent kernel modification to Turnbull's univariate algorithm as a way to fill in the gaps between the resulting innermost intervals. We generalize this approach to the plateau of interval censored and masked competing risks. The new approach is illustrated by applying the method to a human trypanosomiasis data set.

1. Introduction

In this paper, we develop a nonparametric kernel density estimator for modelling interval censored competing risks data. A nonparametric maximum likelihood estimator (NPMLE) for interval censored competing risks data was first proposed by Hudgens *et al.* (2001). Subsequently, Adamic (2006) developed a similar generalization of Turnbull's classical algorithm

AMS: 62N02

for modelling competing risks, but in a more general setting, whereby each failure was allowed to be associated with any subset of possible failure modes (a dynamic known as *masking*). Despite the novelty of these approaches, there was a significant drawback associated with them. As stated in Hudgens *et al.* (2001), these estimators have an unexpected and undesirable property, namely, that the resulting estimator of the survival distribution is undefined over a potentially larger set of regions than the NPMLE of $S(t)$ ignoring failure type. Indeed, the self-consistent, competing risks algorithm or *SC-CR Algorithm* of Adamic (2006) can be shown to converge only over a class of intervals called *cause-specific innermost intervals* – a concept that will be formally developed in this paper. To remedy this phenomenon, we have chosen to generalize a univariate kernel density estimate found in Braun *et al.* (2005) that was used to fill in the gaps between the *innermost intervals* that were created by invoking the self-consistent EM algorithm of Turnbull (1976). The converged estimator of the failure rate distribution is often difficult to smooth, due to the large gaps between innermost intervals as well as the multi-modal shape of the distribution. As mentioned by Duchesne and Stafford (2001), adopting a kernel smoothed estimator at each iteration avoids the bias created by arbitrarily assigning probability mass at the right end points of the innermost intervals and is better at borrowing more information from neighbouring data points than would otherwise be the case. They go on to state that since the innermost intervals are no longer in the picture, the kernel modification moves the algorithm away from problem causing areas that can sometimes get stuck at local solutions.

In this paper, however, a more recent bandwidth selection methodology is employed, based on asymptotic expressions of the mean integrated square

error (MISE) as described in Sheather and Jones (1991) and Wand and Jones (1995). The kernel modification to the SC-CR algorithm is finally applied to a Trypanosomiasis data set, a rare disease found in Africa. The data is subject to interval censoring as well as masking. Interval censoring arises due to the fact that the disease can sometimes remain dormant for many years before symptoms manifest themselves, leaving the time (or age) of infection not precisely known. Masking is present due to the fact that for the two competing risks, being the neurological form and the lymphatic-sanguine form of the disease, the true form cannot always be ascertained accurately.

2. Notation and other preliminaries

This paper assumes that survival data are reported in a case 2 interval censored format, which includes the possibility of exact observations. This type of reporting would be common in any situation where determining the exact time of the event in question would be infeasible, or simply unavailable. A formal definition of the censoring mechanism (taken mostly from Groeneboom & Wellner, 1992, and Schick & Yu, 1998) is given below.

Definition 2.1 *Consider two inspection times U and V , such that $U < V$. Furthermore, assume that the inspection times are independent of the actual failure time X . Then, the intervals $\{L, R\}$ are considered to be a mixture of case 2 interval censored observations with exact observations, if*

$$\{L, R\} = \begin{cases} (-\infty, U], & \text{if } X \leq U \text{ and if } X \text{ is subject to interval censoring} \\ (U, V], & \text{if } U < X \leq V \text{ and if } X \text{ is subject to interval censoring} \\ (V, \infty) & \text{if } X > V \text{ and if } X \text{ is subject to interval censoring} \\ [X, X], & \text{if } X \text{ is not subject to interval censoring} \end{cases}$$

The following standard life table notation, as found in Bowers *et al.* (1997) and Lawless (2003), is applicable to any survival context. First, define ${}_tq_x = \Pr\{\text{failure prior or equal to time } (t + x) \mid \text{subject is currently age } x\}$. For

example, in human mortality studies, $_{10}q_{45}$ would represent the probability a 45-year-old dies in the next 10 years. When working with a life table, it is often convenient to replace q with d to represent the number, instead of the probability, of subjects that fail in a given time interval. Thus, ${}_td_x$ would represent the number of failures in the interval $(x, x + t]$. Also, let ${}_tp_x = \Pr\{\text{survival to time } (t + x) \mid \text{subject is currently age } x\}$. For example, $_{10}p_{45}$ would represent the probability a 45-year-old would survive to age 55. Clearly, it can be seen that ${}_tp_x + {}_tq_x = 1$. When $t = 1$, the subscript is often omitted. Thus, ${}_1p_{45} = p_{45}$, would represent the probability a 45-year-old survives 1 year. As a final example, the probability that a 20-year-old dies between ages 55 and 56 would be written as ${}_{35}p_{20} \cdot q_{55}$.

To motivate the competing risks theory, some additional notation is required. It should also be mentioned at the outset that the terms *competing risk*, *mode of failure*, *decrement*, and *cause* will be used interchangeably, and will be assumed to be mutually synonymous. A superscript in brackets will be used to identify which specific risk is being considered. For example, ${}_tq_x^{(j)}$ would represent the probability of failure for some subject aged x prior or equal to time $x + t$ by cause j in the multiple risk forum. The superscript τ will be used to represent the set of all the risks taken together, in aggregate. For example, ${}_tp_x^{(\tau)}$ would represent the probability of surviving all risks up to time t .

Given the multiple failure mode notation, a generalization for self-consistency pertaining to cumulative incidence functions (CIFs), following the univariate definition and nomenclature found in Duchesne and Stafford (2001), can be offered as follows.

Definition 2.2 *An estimator $\hat{F}^{(j)}(t)$ for a cumulative incidence function is said to be a self-consistent estimator if $\hat{F}^{(j)}(t) = E_{\hat{F}}[F^{(j)}(t) \mid \Psi]$, where*

$F^{(j)}(t)$ is the empirical CIF under complete data, Ψ denotes the set of all of the observed data (complete or censored), and $\hat{F}$ is understood to include the current information in all of the CIFs for modes $1 \dots k$.

The likelihood function for the SC-CR Algorithm can be expressed as,

$$L \propto \prod_{i=1}^n \left[t_{j_i} - t_{j_i-1} q_{t_{j_i-1}}^{(S_i)} \right]^{\delta_i} \left[L_i p_0^{(S_i)} - R_i p_0^{(S_i)} \right]^{1-\delta_i}, \quad (1)$$

where $t_1 < \dots < t_m$ are the distinct values of the observed L_i 's and R_i 's and $t_{j_i} = R_i = L_i$ when $\delta_i = 1$ (corresponding to an exact observation). As well, $S_i \subseteq \{1, 2, \dots, k\}$ represents the masked set of possible failure modes for observation i . For example, ${}_6p_4^{(1,3)}$ would represent the probability that a subject survives causes 1 and 3 between times 4 and 10, given a current age of 4. Based on the notation in the likelihood, 'exact' should be taken here to mean, "since the last time point, t_{j_i-1} ". Equivalently, it will be assumed that $[t_{j_i}, t_{j_i}] \equiv (t_{j_i-1}, t_{j_i}]$, which is consistent with the step-function nature of the nonparametric CIF estimators. Showing that the self-consistent algorithm maximizes (1) is essentially shown in Hudgens *et al.* (2001), and will therefore not be reproduced here.

For ease of appropriation, the steps (and the preamble to the steps before Step 0), statements, and logic of the algorithm will be directly generalized from those of the single variable algorithm as found in the standard survival text of Klein and Moeschberger (1997). First, assume that a total of n subjects are observed as follows: $I_i = (L_i, R_i]$ or $[X_i, X_i]$, $\forall i = 1, \dots, n$, depending on whether I_i is interval censored or an exact observation (the associated masking group, S_i , for observation i is temporarily ignored). Let Δ denote the entire set of time points drawn from $\{L_i, R_i, X_i \forall i\}$, where repeated values should appear only once and all numbers in the set must be finite. Notice that the cardinality of Δ is $\leq 2n$. Then, order all of the

elements of Δ in ascending order. These will now correspond to the time points, $0 = t_0 < t_1 < t_2 < \dots < t_m$, that may be assigned probability masses. The algorithm steps can then be implemented as follows.

SC-CR Algorithm

Step 0 Obtain initial estimates of the probability of each risk occurring at each time point t_r , by evenly distributing the probability mass of each decrement over all of the possible time points and risks included in the observed interval (or exact observation as appropriate).

Step 1 Compute the probability of each risk occurring at time $t_r \forall r$, by dividing the current estimate of the number of failures for each risk at each t_r by the total number of observed intervals (i.e. the number of subjects in the data set, or n). If this is the first iteration, use the results from Step 0.

Step 2 For iteration K , estimate the number of failures for each cause, j , at each time, t_r , using,

$$\hat{d}_r^{(j),K} = \sum_{i=1}^n \left[\frac{\alpha_{ijr} \cdot {}_{t_{r-1}}\hat{p}_0^{(\tau),K} \cdot {}_{t_r-t_{r-1}}\hat{q}_{t_{r-1}}^{(j),K}}{\sum_k \sum_l \alpha_{ikl} \cdot {}_{t_{l-1}}\hat{p}_0^{(\tau),K} \cdot {}_{t_l-t_{l-1}}\hat{q}_{t_{l-1}}^{(\tau),K}} \right] \quad \forall r, l = 1, \dots, m,$$

where

$$\alpha_{ijr} = \begin{cases} 1, & \text{if } X_i \in (L_i, R_i] \text{ could have occurred at } t_r \text{ by cause } j \\ 0, & \text{otherwise.} \end{cases}$$

Note that α_{ijr} automatically takes into account any masking.

Step 3 Compute the estimated number at risk at time t_i using,

$$\hat{Y}_i^K = n - \sum_{f=1}^{i-1} \sum_j \hat{d}_f^{(j),K}.$$

Step 4 Compute ${}_{t_i}\hat{p}_0^{(\tau)}$ and ${}_{t_i-t_{i-1}}\hat{q}_{t_{i-1}}^{(j)} = \hat{f}^{(j)}(t_i)$ say, $\forall i$, using a generalized cause-specific Product-Limit estimator based on the current estimates of $\hat{d}_i^{(j)}$ and $\hat{Y}_i$ (cf. Lawless, 2003, for details). If,

$$\sup_{i,j} \left| \hat{f}_K^{(j)}(t_i) - \hat{f}_{K-1}^{(j)}(t_i) \right| < \epsilon,$$

(jointly $\forall t_i, j$ and given some small predetermined $\epsilon > 0$) stop the algorithm; otherwise return to Step 1.

The SC-CR Algorithm under interval censoring will not converge for all of the potential probability masses, $\hat{f}^{(j)}(t_i)$, at each time point. Analogous to the univariate case, where convergence is reached only over the sample innermost intervals (Turnbull, 1976), the algorithm in the competing risks framework will only converge over *cause-specific innermost intervals*. A formal definition, based on the univariate definitions of Duchesne and Stafford (2001) and Yu *et al.* (1998), is as follows.

Definition 2.3 Let $I_i^{(S_i)}, i = 1, \dots, n$, represent each of the observed failure times, associated with a masking set S_i . Let $J_r, r = 1, \dots, m$, represent all of the disjoint intervals that are non-empty intersections induced by $I_1, \dots, I_n$, ignoring the associated masking set. Then, associate each J_r with each unique cause $j, j = 1, \dots, k$. That is, induce $J_r^{(j)}$ to represent interval J_r associated with cause $j, \forall r, j$. Then, the sample cause-specific innermost intervals will be those cause-specific intervals $J_r^{(j)}$ that satisfy the following criteria:

$$J_r^{(j)} \cap_{int} I_i^{(S_i)} = J_r^{(j)} \text{ and } j \in S_i, \text{ or } J_r^{(j)} \cap_{int} I_i^{(S_i)} \equiv \emptyset \quad \forall r, i, j,$$

where the subscript $\{int\}$ refers to the intervals only, disregarding the associated masking group or failure mode.

As a consequence of this definition, a useful lemma may be stated.

Lemma 2.4 *Exact observations, $[X_i, X_i]^{(S_i)}$, will always produce at least one cause-specific innermost interval for at least one of the causes in S_i at that particular time point t_i .*

As a brief numerical illustration, consider the following data set with two competing risks (death and withdrawal, say), where the second and third failures are partially masked (as opposed to completely masked):

$$\left\{ (0, 3]^{(d,w)}, (2, 5]^{(d)}, [4, 4]^{(w)}, (2, \infty)^{(d,w)}, (6, 7]^{(d,w)} \right\}.$$

Then, the cause-specific innermost intervals are $(2, 3]^{(d)}, [4, 4]^{(w)}, (6, 7]^{(d)}, (6, 7]^{(w)}$.

3. Kernel density estimation for the SC-CR algorithm

3.1 A self-consistent kernel method

Consider the case where data are interval-censored. In such a situation, Stafford (2005), based on the work of Goutis (1997), states that a natural extension of the weight is to define w_i as a conditional expectation, conditional on the observed interval $I_i = (L_i, R_i]$. That is, let,

$$w_i = E \left[\frac{1}{h} \tilde{K} \left(\frac{X_i - x}{h} \right) \mid I_i \right],$$

so that the kernel estimate of f is,

$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^n E \left[\frac{1}{h} \tilde{K} \left(\frac{X_i - x}{h} \right) \mid I_i \right], \quad (2)$$

for a fixed kernel function $\tilde{K}$ (Stafford, 2005). Now, Braun *et al.* (2005) espouse an intriguing way to evaluate Equation (2): use an iterative algorithm such as Turnbull's algorithm, and the kernel density estimate itself to evaluate the conditional expectation at each iteration. Thus, at the K th step of the algorithm, Braun *et al.* (2005) propose that the density estimate be,

$$\hat{f}_K(x) = \frac{1}{n} \sum_{i=1}^n E_{\hat{f}_{K-1}} \left[\frac{1}{h} \tilde{K} \left(\frac{X_i - x}{h} \right) \mid I_i \right]. \quad (3)$$

For the competing risks setting, an expression analogous to Equation (3) can be developed. For modes of failure indexed by $j = 1, \dots, k$, assume independence between the modes of failure. Then, the cause-specific kernel density estimate for iteration K would be expressed as,

$$\hat{f}_K^{(j)}(x) = \frac{1}{n} \sum_{i=1}^n E_{\hat{f}_{K-1}^{(j)}} \left[\frac{1}{h_j} \tilde{K} \left(\frac{X_i^{(j)} - x}{h_j} \right) \mid I_i^{(S_i)} \right], \quad (4)$$

where an optimal bandwidth should be chosen for the failure density of each independent risk j . Although the independence assumption is not required for the SC-CR Algorithm, it is useful here to avoid the complication of having to invoke a multivariate kernel that reflects the joint distribution between the various competing risks.

The following theorem shows that as the kernel bandwidths of each risk tend to zero, the smoothed estimator of the density approaches the self-consistent density. The theorem statement, steps, and logic of the proof are direct generalizations of an analogous univariate proof from Braun *et al.* (2005).

Theorem 3.1 Let $\check{F}_K^{(j)}(x) = \int_{-\infty}^x \check{f}_K^{(j)}(t) dt$ be the kernel smoothed estimate of the corresponding CIF for risk j at the K th iteration with a cause-specific bandwidth h_j , under an interval censoring scheme with the possibility of

masked failure modes. Then, assuming independent kernels for each risk j ,

$$\lim_{h_j \rightarrow 0} \tilde{F}_K^{(j)}(x) = \hat{F}_K^{(j)}(x) \quad \forall x, j, K.$$

Proof The self-consistent CIF is

$$\begin{aligned} \hat{F}_K^{(j)}(x) &= E_{K-1} \left[F^{(j)}(x) \mid \Psi \right] \\ &= E_{K-1} \left[\frac{1}{n} \sum_{i=1}^n I(X_i^{(j)} \leq x) \mid \Psi \right], \end{aligned}$$

since $F^{(j)}(x)$ captures the relative proportion of observations with $X_i \leq x$ and $J = j$, and is also the NPMLE when using complete data. Bringing the expectation into the sum yields,

$$= \frac{1}{n} \sum_{i=1}^n E_{K-1} \left[I(X_i^{(j)} \leq x) \mid I_i^{(S_i)} \right].$$

Now, since the kernel function is a valid probability distribution function that approaches the empirical distribution as the bandwidth tends to 0, and since each competing risk is smoothed with its own unique kernel bandwidth h_j ,

$$\begin{aligned} &= \frac{1}{n} \sum_{i=1}^n E_{K-1} \left[\lim_{h_j \rightarrow 0} \int_{-\infty}^x \frac{1}{h_j} \tilde{K} \left(\frac{X_i^{(j)} - u}{h_j} \right) du \mid I_i^{(S_i)} \right] \\ &= \lim_{h_j \rightarrow 0} \frac{1}{n} \sum_{i=1}^n E_{K-1} \left[\int_{-\infty}^x \frac{1}{h_j} \tilde{K} \left(\frac{X_i^{(j)} - u}{h_j} \right) du \mid I_i^{(S_i)} \right], \end{aligned}$$

since $\int \frac{1}{h_j} \tilde{K} \left(\frac{X_i^{(j)} - u}{h_j} \right) \leq 1$, and therefore the limit can be taken outside of the expectation. By the Fubini-Tonelli theorem, the order of expectation and integration may be interchanged in this case, producing,

$$\begin{aligned} &= \lim_{h_j \rightarrow 0} \int_{-\infty}^x \frac{1}{n} \sum_{i=1}^n E_{K-1} \left[\frac{1}{h_j} \tilde{K} \left(\frac{X_i^{(j)} - u}{h_j} \right) \mid I_i^{(S_i)} \right] du \\ &= \lim_{h_j \rightarrow 0} \int_{-\infty}^x \tilde{f}_K^{(j)}(u) du \\ &= \lim_{h_j \rightarrow 0} \tilde{F}_K^{(j)}(x) \quad \square \end{aligned}$$

3.2 Bandwidth Selection

A key issue when employing any kernel density estimator is the choice of bandwidth. As is well known, the choice of bandwidth is far more crucial than the choice of kernel function, $\tilde{K}$. Also, the magnitude of the bandwidth is always a tradeoff between the bias and variability of the smoothed density and therefore a compromise is always necessary. Braun *et al.* (2005) espoused the use of a cross-validation (CV) scheme, where a cross-validated likelihood was to be maximized based on the Kullback-Leibler distance between f and $\hat{f}_h$. This approach is somewhat awkward to employ in our context, since the elimination of one cause-specific innermost interval would sometimes change the entire support of the failure times, resulting in several missing points. The number of missing points would be different for each innermost interval deleted, which is not consistent with the assumptions used in most CV methods. Indeed, Braun *et al.* (2005) admit that, ultimately, their CV method resembles a form of a k -fold cross-validation technique.

In view of these issues, we have chosen to use a different approach. This approach is based on an asymptotic (AMISE) expansion of the mean integrated square error of the form

$$MISE = \frac{1}{nb} \int \tilde{K}^2 + \frac{1}{4} b^4 \int (f'')^2 \left\{ \int x^2 \tilde{K} \right\}^2 + O\left(\frac{1}{nb} + b^4\right), \quad (5)$$

as described in Venables and Ripley (1997). This method does not suffer from the problems associated with innermost intervals and, as noted by Venables and Ripley (1997), this method is quite close to optimal (in the sense of the bias/variability tradeoff) and preferred to the older cross-validation methods. When the remainder is neglected in Equation (5) the optimal bandwidth can be

shown to be equal to

$$b_{AMISE} = \left[\frac{\int \tilde{K}^2}{n \int (f'')^2 \left\{ \int x^2 \tilde{K} \right\}^2} \right]^{\frac{1}{5}}, \quad (6)$$

as stated in Venables and Ripley (1997). Details for Equations 5 and 6 are given in Sheather and Jones (1991) and Wand and Jones (1995). The S-Plus function *bandwidth.sj*, named after the methodology of Sheather and Jones (1991), uses the *solve-the-equation* method to solve for b in Equation (6) (Venables & Ripley, 1997). The function *bandwidth.sj* was integrated into the kernel-modified SC-CR Algorithm that was programmed in S-Plus. This program was then used to model the trypanosomiasis data.

4. Application

The *Trypanosoma brucei* is a parasite that causes the rare disease *African trypanosomiasis*, colloquially referred to as African sleeping sickness. The parasite is transferred to humans by the bite of the tsetse fly. A person can be infected for long periods without exhibiting any symptoms. Indeed, when symptoms do appear, the disease is often already at an advanced stage of development. There are two distinct forms that the disease can take: the neurological form (N) and the lymphatic-sanguine (LS) form. These comprise the two competing modes of failure, where the failure time, t_i , is the age a person is first infected by the parasite. This diagnosis phase is crucial because it determines the appropriate type of treatment, which is highly toxic.

The data is from the *Bulletin Épidémiologique Hebdomadaire* published by the *Institut de Veille Sanitaire* (Legros *et al.*, 2006), and is summarized in Table 1. There were 26 cases of the disease reported in France during the years 1980 and 2004. Due to the long incubation period where the disease

was not detected, the exact age at infection was not always known precisely, resulting in numerous interval censored observations. Also, in a few cases, the exact form of the disease (N or LS) was not able to be determined. This resulted in observations that contained masked competing risks. The SC-CR Algorithm of Adamic (2006) can therefore be used to model the cumulative incidence failure times under these two forms of the Trypanosomiasis disease. The SC-CR Algorithm was run on the data, assuming an epsilon of 0.0000001. The results are given in the second and third columns of Table 2.

Note that less than 50% of the possible data points have cause-specific innermost intervals for the two competing risks. Smoothing these two vectors of probability mass would not be advisable since there are so many gaps between the innermost intervals, making the probability landscape sparse and multi-modal. Use of the kernel modification at each iteration of the SC-CR Algorithm is therefore warranted.

To implement the smoothing modification, the kernel function that will be used is the Parzen function (Parzen, 1962). Independent kernels for each risk seemed justified, since the correlation coefficient of the two competing densities before smoothing was only -0.07216. Thus, a multivariate kernel was not necessary in this case. Again, assuming an epsilon of 0.0000001, and utilizing the optimal bandwidth selection method described in Section 3.2 at each iteration, convergence was achieved after 61 iterations. These results are also shown in Table 2.

The kernel modification produced revealing results. Note that all of the gaps between the innermost intervals were successfully filled in an optimal manner. In turn, this produces valuable information that was not previously available. First, there are obviously non-zero probabilities of decrement at each of the possible ages in reality.

Table 1. Summary of the Trypanosomiasis data

Case	Year	Age at infection	Form of Disease
1	1980	[20,20]	N
2	1981	(0,29]	LS
3	1981	[29,29]	LS
4	1981	(0,32]	LS
5	1981	[43,43]	N
6	1982	(0,19]	LS
7	1982	[23,23]	N
8	1983	(0,40]	LS or N
9	1984	[24,24]	N
10	1984	(0,31]	N
11	1986	[13,13]	N
12	1988	[2,2]	N
13	1988	(0,34]	N
14	1990	[27,27]	N
15	1991	[34,34]	N
16	1992	(0,62]	LS or N
17	1992	(0,50]	N
18	1993	(0,30]	N
19	1993	[27,27]	LS
20	1993	[30,30]	N
21	1996	[53,53]	N
22	1997	(0,53]	LS
23	1998	[45,45]	LS
24	1999	(0,50]	LS
25	1999	(0,34]	LS or N
26	2001	[28,28]	N

Table 2. Converged failure rates for the Trypanosomiasis data set

t_i	$\hat{f}^{(N)}(t_i)$	$\hat{f}^{(LS)}(t_i)$	$\check{f}^{(N)}(t_i)$	$\check{f}^{(LS)}(t_i)$
2	0.0647612	0.0328146	0.0390393	0.0249532
13	0.0647612	0.0328146	0.0439835	0.0361362
19	0.0000000	0.0328159	0.0424939	0.0273766
20	0.0647612	0.0000000	0.0506859	0.0059601
23	0.0647612	0.0000000	0.0611600	0.0026659
24	0.0647612	0.0000000	0.0643542	0.0160377
27	0.0647612	0.0984450	0.0609008	0.0678292
28	0.0647612	0.0000000	0.0496266	0.0313408
29	0.0000000	0.0984450	0.0374755	0.0670994
30	0.0647612	0.0000000	0.0291221	0.0128623
31	0.0000000	0.0000000	0.0195978	0.0009991
32	0.0000000	0.0000000	0.0173620	0.0000779
34	0.0518090	0.0000000	0.0220405	0.0000137
40	0.0000000	0.0000000	0.0228129	0.0002045
43	0.0427938	0.0000000	0.0196209	0.0060269
45	0.0000000	0.0519712	0.0131585	0.0396535
50	0.0000000	0.0000000	0.0127404	0.0060224
53	0.0400000	0.0000000	0.0169662	0.0001745
62	0.0000000	0.0000000	0.0143953	0.0000029

This is immediately rectified once the kernel modification is employed. Secondly, the magnitudes of the failure probabilities are themselves revealing. For example, consider age 40. Before the kernel modification, it was not known whether failure due to cause N or cause LS was more likely (since they were both zero). Now we know that infection at age 40 is over 100 times more likely to be due to form N than form LS. Figure 1 depicts the final CIF curves for each of the two competing risks. We note that the light smoothing that is evident in the plot was the result from applying the optimal MISE criterion for this data set.

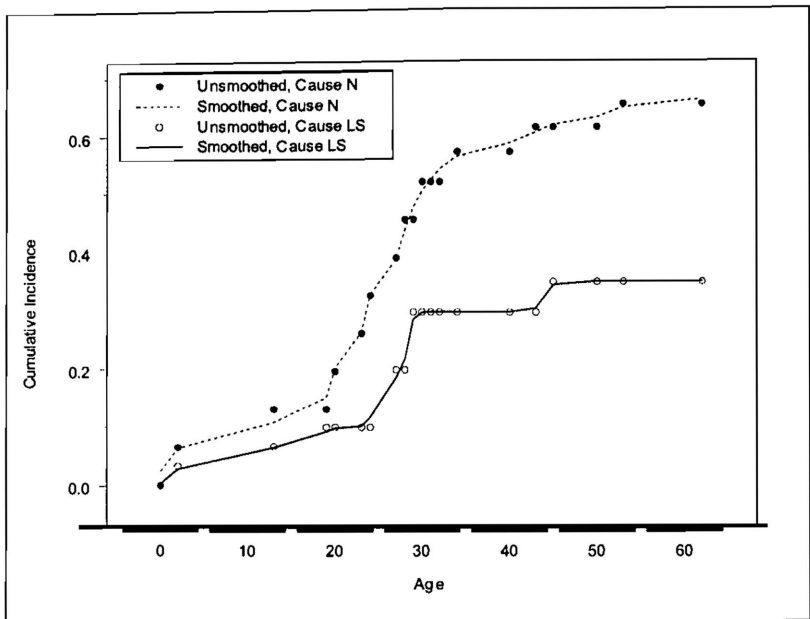


Figure 1. Converged CIFs for the Trypanosomiasis data

5. Discussion

There are other advantages that can be realized from using the kernel modification to the SC-CR Algorithm. First, it is a well established fact that Turnbull's algorithm, like other self-consistent algorithms, can sometimes get stuck at local solutions and not converge to the NPMLE. As described in Day (2005), the Kuhn-Tucker conditions from optimization theory can be used to specify starting conditions that will guarantee convergence. However, as noted by Braun *et al.* (2005), the kernel implementation at each iteration moves the algorithm away from these problem causing areas, making the probability of convergence extremely high and making the need to compute the Kuhn-Tucker conditions minimal. Secondly, the use of the kernel modification tends

to reduce the number of iterations required for convergence. This fact was established in the univariate setting by Braun *et al.* (2005) via simulation. Although we do not present a rigorous proof of this claim here, we also noticed the same phenomenon in the multivariate, competing risks domain, by running the kernel modification under many different scenarios. Finally, we note that the issue of larger bias introduced from arbitrarily placing probability mass at the right end of the innermost intervals, a dynamic outlined in Duchesne and Stafford (2001), is even more relevant in the competing risks framework due to the larger number of innermost intervals involved. As with the univariate kernel method, the kernel modification to the SC-CR algorithm is unarbitrarily data-driven (Braun *et al.*, 2005), thereby making the cause-specific innermost intervals, and the problems they engender, irrelevant, as was also stated in Braun *et al.* (2005).

In terms of future work directly related to the kernel modification, we note that the modification can be applied, not only to interval-censored failures, but to other censoring mechanisms as well. It can be applied to a variety of self-consistent models that include left-censored, doubly-censored, truncated, and other more elaborate censoring schemes that are very much at the forefront of contemporary scholarship in survival analysis.

Acknowledgements

We wish to thank Radu Craiu (University of Toronto, Canada) for helpful correspondence. This research was partially funded by a Laurentian University Research Fund Grant.

References

- ADAMIC, P. (2006). *Self-Consistent Density Estimation Methods for Competing Risks with Group Masked Double and Interval Censored Failures*. Ph.D. thesis, University of Guelph, Canada.
- BOWERS, N.L., GERBER, H.U., HICKMAN, J.C, JONES, D.A., NESBITT, C.J. (1997). *Actuarial Mathematics, 2nd ed.* Schaumburg, Illinois: The Society of Actuaries.
- BRAUN, J., DUCHESNE, T., STAFFORD, J.E. (2005). Local likelihood density estimation for interval censored data. *The Canadian Journal of Statistics* **33**, no. 1: 39–60.
- DAY, B. (2005). Distribution-Free Estimation with interval censored Contingent Valuation Data: Troubles with Turnbull? Available online at http://www.uea.ac.uk/env/cserge/pub/wp/edm/edm_2005_07.pdf.
- DUCHESNE, T., STAFFORD, J.E. (2001). A kernel density estimate for interval censored data. *Technical Report No. 0106*: University of Toronto.
- GOUTIS, C. (1997). Nonparametric estimation of a mixing density via the kernel method. *Journal of the American Statistical Association*, **92**: 1445–1450.
- GROENBPOOM P., WELLNER, J. (1992). *Information Bounds and Nonparametric Maximum Likelihood Estimation*. Basel: Birkhauser.
- HUDGENS, M.G., SATTEN, G.A., LONGINI Jr., I.M. (2001). Nonparametric Maximum Likelihood Estimation for Competing Risks Survival Data Subject to Interval Censoring and Truncation. *Biometrics*, **57**: 74–80.
- KLEIN, J.P., MOESCHBERGER, M.L. (1997). *Survival Analysis: Techniques for Censored and Truncated Data*. New York: Springer.
- LAWLESS, J.F. (2003). *Statistical Models and Methods for Lifetime Data, 2nd ed.* New York: John Wiley and Sons.
- LEGROS, F., ANCELLE, T., et le réseau Anofel, (2006). Trypanosomiase humaine africaine: recensement des cas d'importation observés en France, 1980–2004. *Bulletin épidémiologique hebdomadaire* **7**: 57–59.

- PARZEN, E. (1962). On estimation of a probability density function and mode. *Ann. Math. Stat.* **33**: 1065–1076.
- SCHICK, A., YU, Q. (2000). Consistency of the GMLE with Mixed Case Interval-Censored Data. *Scandinavian Journal of Statistics* **27**: 45–55.
- SHEATHER, S.J., JONES, M.C. (1991). A reliable data-based bandwidth selection method for kernel density estimation. *Journal of the Royal Statistical Society, Series B* **53**: 683–690.
- STAFFORD, J. (2005). ‘Exact’ Local Likelihood Estimators. Available online at <http://www.gla.ac.uk/external/RSS/RSScomp/Stafford.pdf>.
- TURNBULL, B.W. (1976). The empirical distribution function with arbitrarily grouped, censored and truncated data. *Journal of the Royal Statistical Society* **38**: 290–295.
- WAND, M.P., JONES, M.C. (1995). *Kernel Smoothing*. Chapman & Hall.
- YU, Q., LI, L. & WONG, G.Y.C. (1998). Asymptotic variance of the GMLE of a survival function with interval-censored data. *Sankhya: The Indian Journal of Statistics*, Vol. **60**, Series A, Pt. 2, pp. 184–197.